Information Extraction From Fiscal Documents Using LLMs

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Abstract

Large Language Models (LLMs) have demonstrated remarkable capabilities in text comprehension, but their ability to process complex, hierarchical tabular data remains underdeveloped. We present a novel approach to extracting structured data from multi-page government fiscal documents using LLM-based techniques. Applied to large fiscal documents from the State of Karnataka in India, our method achieves high accuracy by combining domain knowledge, sequential context, and algorithmic validation. Traditional OCR methods work poorly to translate information from PDF documents into a structured database of fiscal information, with errors that are hard to detect. The inherent structure within fiscal documents, with totals at each level of the hierarchy, opens up the possibility to extract information, with robust internal validation of the extracted data. We demonstrate that LLMs can read tables and also process document-specific structural hierarchies, offering a process for converting PDF-based fiscal disclosures into research-ready databases that can be scaled up. We use the hierarchical relationships within fiscal documents to create multi-level validation checks. Our implementation shows promise for broader applications involving complex tables, particularly for developing country contexts.

Keywords

LLMs, tabular data, information extraction

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1 Introduction

While LLMs excel at processing natural language, their ability to process tabular data has not been as emphasised or understood. In the real world, a great deal of financial disclosures, bills, and fiscal documents are published in PDF formats containing tables of data that have inherent, complex relationships. These are non-linear in the flow of information, that are not easily machine-analysable. While the source information exists in the PDF, data analysis requires human knowledge to understand the document structure and parse it correctly. A simple extraction of information from PDF files is particularly prone to errors when tables have inconsistent presentation of text and numbers. Research in this field is hampered by the lack of a large corpus of PDF documents and their associated parsed, structured text.

In India, complex hierarchical public finance databases are released as large PDF files. While the PDF files contain all the information and are readily available, many applications require the underlying database, and this is hard to extract. In this paper, we show an LLM-based process that is able to map the public domain PDF files into accurate datasets. While the ideas are general, our implementation is done using the fiscal documents of the Indian state of Karnataka. For this setting:

- We show that the format of information in the fiscal documentation of Karnataka is representative of the complexity of fiscal disclosures of Indian states.
- We develop a process for extracting machine-analysable information from the fiscal documents of this state.
- We demonstrate how to identify the accuracy of the information extracted, even when no ground truth dataset exists. Relying on the structure of fiscal documents, our mechanism also identifies the locations where the information extraction failed.

1.1 The Challenge

Sophisticated applications of public finance data require analysis applied to the underlying dataset. Economists, civil society organizations and journalists can use this data to tag different heads of expenditure, and categorize spending by function, geography, department, and other dimensions. This allows for an analysis of government spending patterns, efficiency, and shifts in priorities over time. Examples of applications include:

- Public procurement analysis
- Budgeting efficiency studies (Budgeted vs Revised vs Actuals)
- Expenditure priority tracking over time
- In-year fund reallocation analysis
- Functional/departmental/geographic classification

When the public finance dataset is represented as a PDF file, this hinders analysis. Traditional OCR approaches are not effective in recovering the underlying data because:

- Tables span multiple pages; tables have varying structures.
- The files are large: roughly hundreds of pages. Since tables span multiple pages, page-by-page extraction missed vital information.
- In India, the documents feature an inconsistent mix of English and regional languages.
- Units and formatting vary across documents.

For researchers and civil society organisations aiming to analyse public finance, this creates a significant barrier. Manual data entry is labor-intensive and error-prone. Converting these documents into structured, machine-readable formats through a mix of text and image processing tools is a laborious task that requires manual intervention and verification, as well as domain expertise with public finance. The need for such effort acts as a barrier to scaling up both the availability of structured databases of fiscal information, as well as the analysis that is possible when such large scale databases are available.

1.2 The Opportunity

Unlike general text extraction, the deep hierarchical fiscal documents contain a natural validation mechanism: columns must sum to reported totals at upper hierarchical levels. This creates a unique opportunity to develop verifiable LLM-based extraction systems where extraction accuracy can be algorithmically validated.

It is a big step forward for data extraction when structured data is extracted with accuracy scores and identification of the precise locations of errors. Downstream users could potentially expend a small burden of additional effort to solve the last mile of cleaning, and then arrive at a perfect dataset.

This work may have important implications for future refinement of LLMs themselves. LLM performance on multi-lingual, hierarchical, multi-page tabular data is known to be poor. Our research program can yield a real world corpus of tens of thousands of pages of parallel PDF and structured-text data, which can be used for training LLMs. When such a dataset is created, it will induce improvements in LLMs for processing such tabular information.

2 LLM approaches

LLMs have been used for data extraction and data summarisation, and for Q&A tasks. Early work by Dong et al. [2] covered common formats like web documents, spreadsheets and CSVs, but PDFs were either under represented, or entirely missing. A recent survey on small, English language tables from Wikipedia by Sui et al. [6] finds that LLMs do not have a good understanding of table structure. Currently, most research LLMs focuses on understanding tabular data primarily for the question answering task. Most papers (ex. Zhao et al.[8], Bhandari et al. [1]) focus on machine readable formats like JSON, or text formats delimited by pipe characters. Even working with simpler formats, Zhao et al. [8] demonstrate that tabular data with hierarchical information is poorly understood by current LLMs. Lu et al. [3] published a survey of LLM techniques showing that academic research has been focused on easier, text-based document types (text, JSON, HTML), and has not fully investigated image, or page-based formats like PDF, and has not fully explored real-world complexity: multi-page tables, multi-lingual tables, hierarchical tables, and PDF files spanning multiple pages. The authors conclude, "*Future table LLMs should adapt quickly and cheaply to real world business needs. Research directions include synthesizing high-quality training data that reflects the diverse needs of specific domains by cost-effective methods.*"

There have been efforts to create evaluation datasets (ex. [4, 7, 10]) with complicated tables representative of real-world use-cases. At its most complex, the TableEval [10] dataset has dual lingual (English and Chinese), hierarchical tables generated from computer spreadsheets provided as JSON or HTML tables. The largest table in that dataset (table 520) has 113 rows and 3 columns, and is a 10 kilobyte spreadsheet. TableEval is intended for Q&A tasks rather than direct information extraction.

In contrast, the real world problems that we face are much more complex. The tables in Indian public finance data have evolved over multiple decades, and are published in the public domain as PDF files, each containing hundreds of pages. There are many different tables of varying layouts, and tables can spread across pages and contain hundreds of rows. They have interspersed English and Indic characters with inconsistent coding.

3 Methodology

3.1 Key Innovations

While straightforward prompting can work for simple tables, complex hierarchical tables like fiscal documents require additional techniques to ensure accuracy and consistency. To ensure high-quality extraction in a format that is also algorithmically verifiable, we introduce the following innovations:

(1) **Image-based processing**: Converting PDF pages to high-resolution JPGs (300 DPI) improves LLM comprehension compared to direct PDF input or text-based OCR. Converting a document to image removes all text metadata, which then forces the LLM to perform OCR to recognize text terms, rather than reading the (potentially erroneous) text metadata. The LLM may also utilise information represented in the physical placement of rows and columns in a table. At the same time, working with image files for each page introduces

new complexities where the context from previous pages needs to be carried forward manually.
(2) **Sequential context**: The PDF files that we face have hundreds of pages, and rapidly run into the limited context window. Hence, each page receives the previous page's extracted data as context, enabling state carry-forward across page boundaries.
(3) **Multi-level validation**: Algorithmic checks verify summation consistency at the various levels of fiscal reporting: Object Head, Detailed Head, Sub Head, Minor Head, Sub Major Head and Major Head levels.
(4) **Meta-prompting**: Our initial prompt provides domain context about fiscal documents, and provide few-shot examples to the LLM, to get the LLM to write the extraction prompt.
(5) **Intelligent cleaning**: A semantic CSV cleaner uses row-type understanding (Header/Data/Total) to detect and correct column misalignment.

4 Implementation: Karnataka Finances 2020-21

4.1 Precise Problem Statement

The input PDF files have the following features:

(1) **Multi-lingual source**: India is a vast, diverse country with 22 official languages written in different scripts. States conduct business in English, in addition to an official state language. Fiscal information is usually published in English, but can also carry the state's official language. In some cases, the source documents are only published in the state's language, and not in English. Further, Indian languages are "low-resource" languages, ie, they suffer from lack of a high quality corpus on the web. This leads to worse LLM performance on both information extraction and generation tasks, e.g. as demonstrated by Singh et. al [5].
(2) **Document Encoding**: Since the documents are published as PDFs, they can carry font information. We find that the regional language is often encoded incorrectly, or is encoded as ASCII and font code books are used to display the regional language. Information extraction from such malformed documents is difficult, as machine extraction "sees" ASCII in the metadata. This is true of both conventional metadata-extraction tools (pdf2text) and language models. Hence, working with image files is essential, which has its own consequences.
(3) **Document Size**: PDF files with more than 500 pages are common. As an example, Table 2 on page 5 shows the number of pages in the PDF of files that we have analyzed. An information extraction technique has to handle this immense size. As mentioned above, a naïve approach involving LLMs chokes on overflowing the context window.
(4) **Table Structure**: The source documents contain tables of different types. The initial tables contain the top-level hierarchy, and overall totals. Subsequent tables expand on this with department-level detail. LLM prompts need to explain this structure.
(5) **Table Locations**: Fiscal tables often span many pages. Page headers and footers have to be ignored, and the extraction process needs to remember where a table starts and ends. Further, pages can contain multiple tables, as a single page can end a large table from prior pages, fully contain one or more small tables, and start another table that continues for many pages. Any single-page extraction prompt needs to consider this level of complexity. While most pages are in portrait orientation, sometimes wide tables are rotated, and so intervening pages are in landscape orientation.
(6) **Non-standard Number Formatting**: The files use lakhs (10^5), crores (10^7), thousands, mixed use of commas and periods, and sometimes non-Arabic numerals.
(7) **Scanned Documents**: Some documents are scanned images, others are digitally generated PDFs.
(8) **Hierarchical Tables**: There are multiple nested hierarchies, and not all levels are consistent within or across states.
(9) **Inconsistent Formatting**: Merged cells, inconsistent spacing, additional textual context in the middle of tables that may require semantic understanding to parse correctly.
(10) **Repeated Values**: Similar numbers repeated in different categories (Revenue / Capital, Voted / Charged).

Each Indian state releases public finance data through PDF files which have some subset of these 10 elements of challenge. The state of Karnataka is a particularly difficult problem, featuring 8 out of 10 of these challenges (i.e. everything enumerated above other than problems 6 and 7).

4.2 Approach

Our method works in the following steps:

(1) **Document structure** We provide the LLM with knowledge about the accounting structures specific to Indian government fiscal documents, including definitions of Major Head, Minor Head, Object Head, and the typical hierarchical relationships. This hierarchy relationship is visualised in Figure 1 on page 4. The LLM is also provided the full PDF document to get the various table types and nesting hierarchies.
(2) **Table structure** We take the document structure, and prompt the LLM to create CSV schema for each type of table, specifying column names, data types, and expected formats for numerical and categorical data.
(3) **Meta-prompt strategy** We give the LLM illustrative pages from the document, with the CSV schema from Step 2, and ask it to generate the actual extraction prompt.

This prompt is then used to extract each page of the document sequentially, passing the extracted data from the previous page as context for the current page. For each page, the LLM identifies the table type, applies the relevant schema, and extracts the data into that specific CSV format. If there are multiple table types on a single page, the LLM segments the page accordingly and applies the appropriate schema to each segment.

4.3 Output Schema

For Karnataka, the LLM generates five CSV archetypes:

(1) Sub-Major Head
(2) Minor Head
(3) Sub Head
(4) Detailed Head

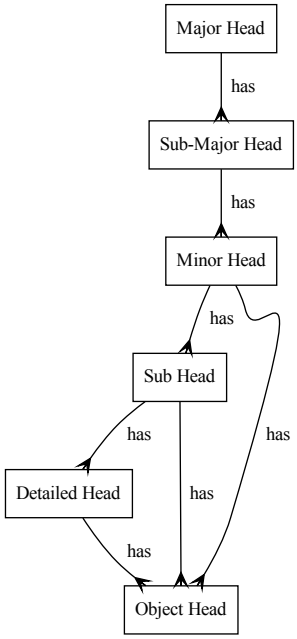


Figure 1: Hierarchy in Karnataka finances

Vol.	Validation Type	Checks	Passed	Pass Rate %
1	Object → Detailed Head	374	317	85%
	Two-source Minor Head	154	146	95%
	Overall	528	463	88%
2	Object → Detailed Head	337	284	84%
	Two-source Minor Head	126	118	94%
	Overall	463	402	87%
3	Object → Detailed Head	199	149	75%
	Two-source Minor Head	90	84	93%
	Overall	289	233	81%
4	Object → Detailed Head	189	147	78%
	Two-source Minor Head	60	59	98%
	Overall	249	206	83%
5	Object → Detailed Head	275	203	74%
	Two-source Minor Head	115	113	98%
	Overall	390	316	81%
6	Object → Detailed Head	99	79	80%
	Two-source Minor Head	56	55	98%
	Overall	155	134	86%
7	Object → Detailed Head	161	125	78%
	Two-source Minor Head	76	73	96%
	Overall	237	198	84%
	All Volumes	2311	1952	84%

Table 1: Validation results

(5) Object Head

4.4 Cleaning

The raw LLM output contains misalignment due to merged cells, inconsistent spacing, and occasional misclassifications of row types. We implement a semantic CSV cleaner that uses the understanding of row types (Header/Data/Total) and row levels to detect and correct column misalignment. This cleaner ensures that each row adheres to the expected schema, correcting for common errors such as shifted columns or misplaced totals.

4.5 Validation Results

Since we do not have the ground truth, we perform two types of internal consistency checks as a validation of the extraction process:

- **Numerical Consistency**: By verifying sums of Object Heads against corresponding Detailed Head totals, we ensure within-schema consistency. By matching Minor Head totals from two different schema we ensure across-schema consistency. It is possible to perform the within- and across- schema numerical consistency checks at all levels of hierarchy.
- **Structural Soundness**: We calculate the Tree Edit Distance Similarity (TEDS), an established metric from Zhong et al. [9] to compare hierarchical structure. TEDS measures how accurately fiscal heads across different schema match identical structures at the same depth. Here, the numbers are irrelevant, and we're looking to match the hierarchical structure from various sources to validate the extraction process.

Table 1 presents the results for **numerical consistency** checks for all seven volumes of the Karnataka state finances in Year 2020–2021. Due to the multiple independent sources of numerical validation, errors in consistency checks can be traced back to specific pages and rows, allowing for targeted manual review or re-extraction. Our pipeline prints the locations where the specific failures occurred.

As an example, the following are the locations of failures from the Object → Detailed report on Volume 1:

```
Major_Head              2039
Description             Total 09
Page                    3
Status                  FAIL
Accounts_2018_19_Match  PASS
Budget_2019_20_Match    FAIL
Revised_2019_20_Match   FAIL
Budget_2020_21_Match    PASS
```

Table 2 presents results for our implementation of **structure soundness** through the Tree Edit Distance-based Similarity (TEDS) measure from the paper by Zhong et al. [9]. For each of the five extracted schema within a volume, tree structures were created for each Major Head hierarchy. The tree structures for two schema at the same depth were then compared. Since the two schema are getting generated from different pages in the PDF document, we effectively cross-check fiscal-head hierarchies from two different locations in the same source. A TEDS score of zero implies that the structures are identical. We divide the number of instances with zero TEDS against all calculated TEDS for each volume and

report that as the single accuracy number in Table 2. Across all the volumes of the Karnataka government, we have been able to achieve an accuracy of 73% to 96%.

File	Pages	Accuracy
Volume 1	227	95.24%
Volume 2	179	73.68%
Volume 3	139	88.00%
Volume 4	142	83.33%
Volume 5	181	96.77%
Volume 6	74	91.30%
Volume 7	112	79.17%

Table 2: Tree Edit Distance Similarity (TEDS)

We summarise the successful features of our work as three key ideas:

(1) The ability to get accuracy metrics in the absence of ground truth data.
(2) The ability to identify locations where the extracted data does not match the original PDF. Errors at these locations can then be corrected with hand-labeling. Sometimes, the extracted data can be used if failures occur in locations that are not under analysis.
(3) The understanding of multi-level hierarchy during extraction, and corresponding validation of the hierarchy through different sources in the same PDF document.

4.6 Computation Costs

We used Gemini 2.5 Pro for the extraction. Table 3 presents the token count for all volumes of Karnataka's 2020-21 finances to establish a computational baseline for replication. The breakdown of input, thought, and output tokens provides a granular view of the resource allocation required for this pipeline.

File	Pages	Input Tok.	Thought Tok.	Output Tok.
Volume 1	227	1567223	1898941	443077
Volume 2	179	1412790	1486710	430499
Volume 3	139	977091	1140116	299388
Volume 4	142	1052354	1207653	317168
Volume 5	181	1452697	1499687	421164
Volume 6	74	466169	583537	123694
Volume 7	112	842688	1010741	235464

Table 3: Gemini 2.5 Pro Token count

5 Future Directions

We have demonstrated a technique for extracting information from state fiscal documents by using a single Indian state (Karnataka) as an example. Our research goal is to extend this work across all Indian states and over many years of fiscal disclosures, to create a trusted database about Indian state finance. This will require many improvements in the implementation:

- **Automated meta-prompting**: Develop a pipeline where the LLM:
  - Analyzes document structure (table types, page ranges).
  - Generates CSV schema per table type.
  - Creates extraction prompts automatically.
  - Cleans up the output using a mix of rule-based and LLM-based cleaning.
  - Validates and iterates until the validation passes.
- **Cross-state robustness**: Test across multiple Indian states with varying document structures.
- **Multi-year consistency**: Ensure format stability across fiscal years.
- **International applicability**: Extend to fiscal documents from other developing economies.
- Create large scale reference datasets of table databases that can feed back into AI training for the future.

5.1 Reproducible Research

All materials, including the fiscal documents, source code, validation scripts, extracted information and instructions are available under MIT license at: https://github.com/xKDR/LLM_table2db.

6 Conclusion

We have shown a path for LLMs to read and understand complex hierarchical tables when provided with appropriate domain knowledge, sequential context, and validation mechanisms. Our approach understands the structure of 200+ page PDF documents, and extracts it into research-ready machine-readable files with high accuracy. Our mechanism also provides the locations of data extraction failures, even though the ground truth is not observed.

These results suggest that LLMs do much more than regular expression matching. They are able to leverage their vast domain knowledge acquired in pre-training, so that prompts are able to steer them effectively. Combined with built-in validation through summation checks, this enables reliable automation of a task previously requiring extensive manual effort.

While it has been applied here to Indian state finances, the methodology is broadly applicable where complex tabular data is published in non-machine-readable formats and where internal consistency checks can be defined. This opens up new avenues for leveraging LLMs for extracting datasets from PDF files in many situations.

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